Cx1106
Computer Organization and Architecture

Overall Review

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Review of Part 2

- Nature of Part 2
  - Diversified scope of content
  - Many types of devices covered
    - Heavy content
    - Some described only at high-level
- management of content
  - Subject to exam
    - Short answered questions
    - MCQs

    Working examples/explanations (more to come)
  - Not subject to exam
  - Platforms for students with questions: Q/A sessions, on site/line, emails, Teams

Recap of Selected Major Topics

- Processor Core & Computer Arithmetic
  - Pipeline
  - Pipeline conflicts
  - Positional Numbering System
  - Carry & Overflow
  - IEEE 754 Floating Point Standard
  - consideration during coding
- Signal Chain Sub-System
  - ADC-DAC
  - Parallel and Serial (UART/SPI) Digital Interfaces
  - Polling
  - Direct Memory Access Controller
  - Interrupt Controller
  - Comparisons of different interfaces and methods

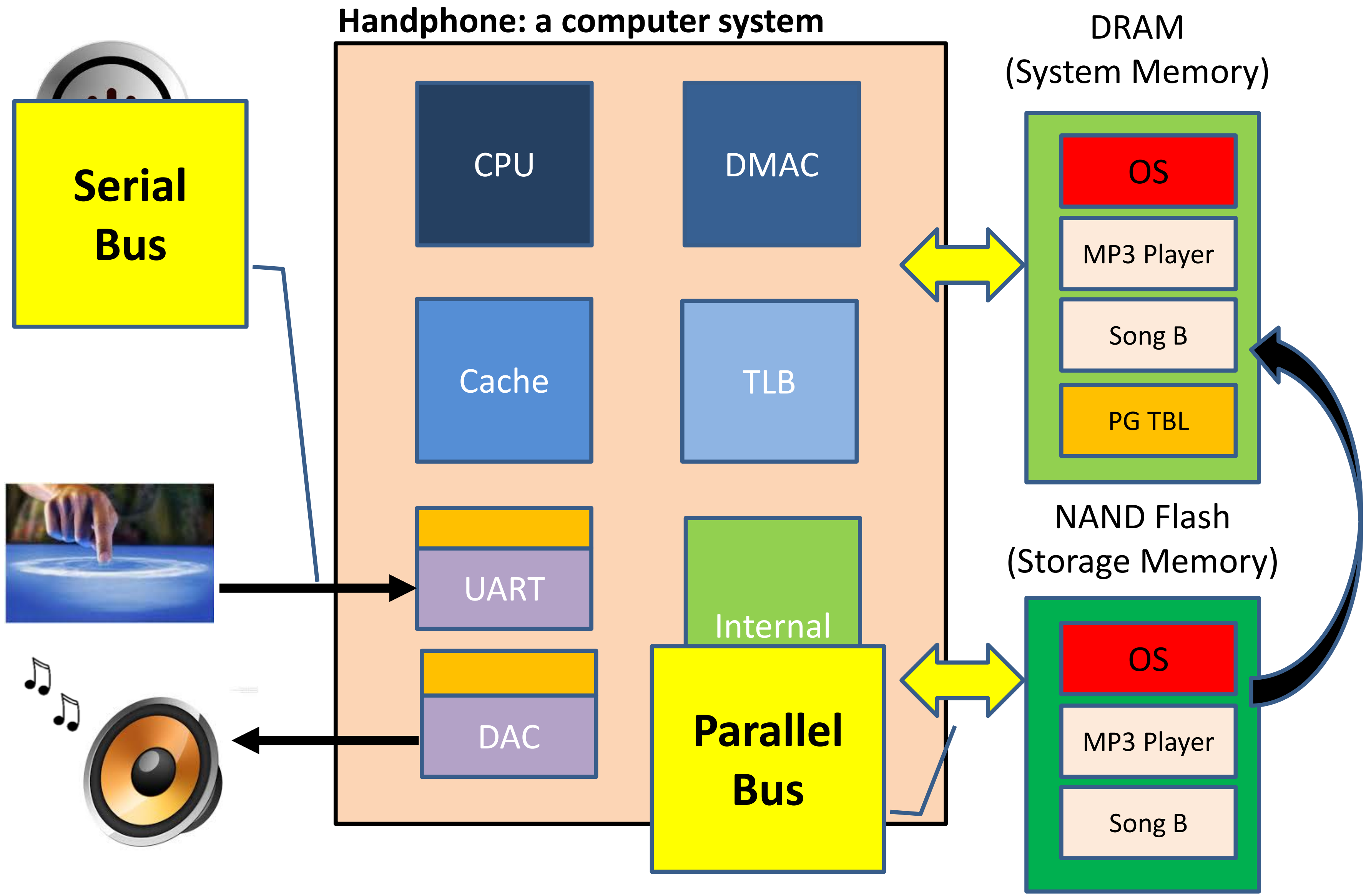
Recap of Selected Major Topics

- Memory Sub-System
  - Semiconductor Memories (SRAM, DRAM, FLASH)
  - Flash memory based Solid State Drives
  - Magnetic Hard Disk Drives
  - Comparisons of different types of memory
  - Cache Memory Management
  - Virtual Memory Management
  - Integrating Cache and Virtual Memory
- Communication and User Interface Sub-System (MCQs only in final exam)
  - Wired (USB, HDMI)
  - Wireless (Wifi, Bluetooth)
  - Input Devices (Keyboard/Mouse, Capacitive Touch, Camera, Microphone)
  - Output Devices (Display, Speakers)

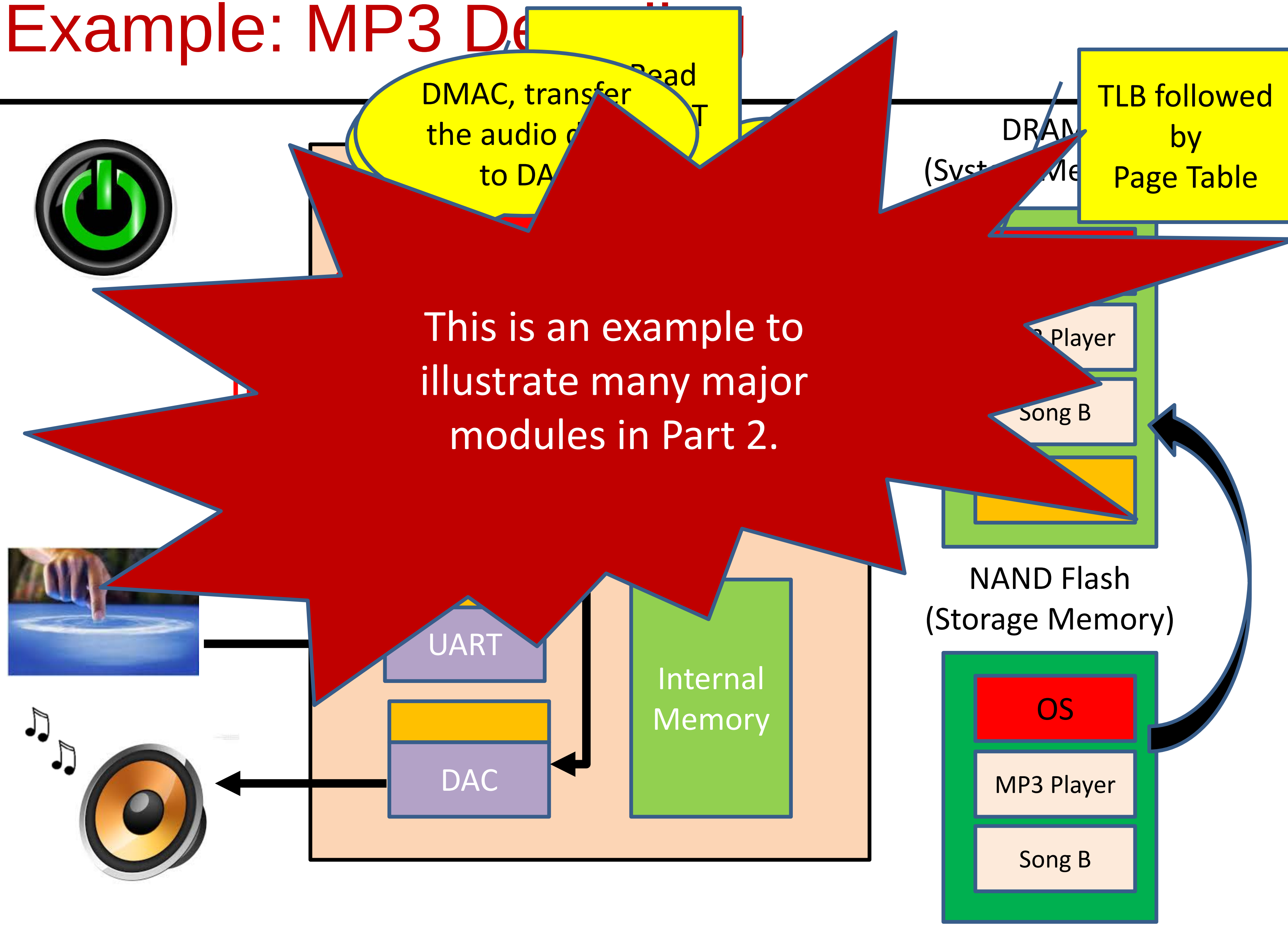
- Additional text description added to lecture slides (as note pages) for the content, under an additional folder of “lec sides with additional notes” in course site, as supplementary material for your revision when needed

- SC1006-Common Terms and Definitions.pdf: also helpful to you

Example: MP3 Decoding in a handphone



Example: MP3 De



OPTIONAL TUT QUESTIONS FOR FURTHER PRACTICE

Optional Questions—Parallel Data Transfer

What is Signal Skew and Crosstalk? Briefly discuss how they affect the maximum transfer speed in an electrical circuit.

- Signal Skew (including clock skew) is the variation in **propagation** speed as electrical signals transfer through the circuit.
  - signals may arrive at slightly different time at the receiving I/O pin(s). If the skew is too large, wrong data will be latched in.
  - **As transfer rate increases, the margin of error for skew decreases**. **This limits the maximum transfer rate in a circuit.**

- Crosstalk is the electrical/electromagnetic **interference** one circuit has over another.
  - Crosstalk increases the noise level in a circuit and affects the accuracy of data received.
  - High speed transfer is more affected as a noise glitch can easily be mistaken as a valid logic transition.
  - **Interference also gets amplified when the transfer rate increase** (more signal transitions leads to increase level of noise and harmonics).

Tut 5

3) There has been a trend of 'serialization' of interface bus standard. E.g. USB replacing Parallel Port Interface, SATA replacing IDE in HDD.

(a) What are the advantages that Serial Bus has over Parallel Bus interface that enticed industry player to move in this direction?

(b) What are the scenarios in which Parallel bus will still be preferred over Serial bus? What are the design considerations that need to be put in place in such cases?

(a) Serial Bus has *less* data lines compared to Parallel Bus.
-> **Less issue** with **signal skew** and less probability of being influenced by **crosstalk**.
-> **Less bulky** so easier to route PCB trace and cabling (SATA vs IDE Cable).
-> Even though data is transferred one bit at a time, modern serial standards are able to achieve relatively high data transfer speed (USB3.0 raw speed is up to 5Gbps). This is achieved through good design and testing conformance/certifications.

(b) Parallel bus is still commonly used when sustained high speed is needed. E.g. interfacing to DDR SDRAM (DDR3 SDRAM data rate scales up to beyond 20GByte/s). But designing these high speed parallel buses require deep knowledge of electrical circuit behaviour, so as to reduce the effect of signal skew and crosstalk. And typically, the length of the parallel bus is very short (order of **cm** rather than metre), with lots of consideration on **grounding** and **shielding**. High speed circuit design is almost an art.

4) UART

- Example:
  - Receiving device is sampling at 115,200 Baud.
  - If the **transmitting** device is sending **0x0D** at 38,400 baud (1/3 of 115,200 Baud), the data received will be **0x1C**.

Receiver Baud Rate = 115200

(LSB first)

Transmitter Baud Rate

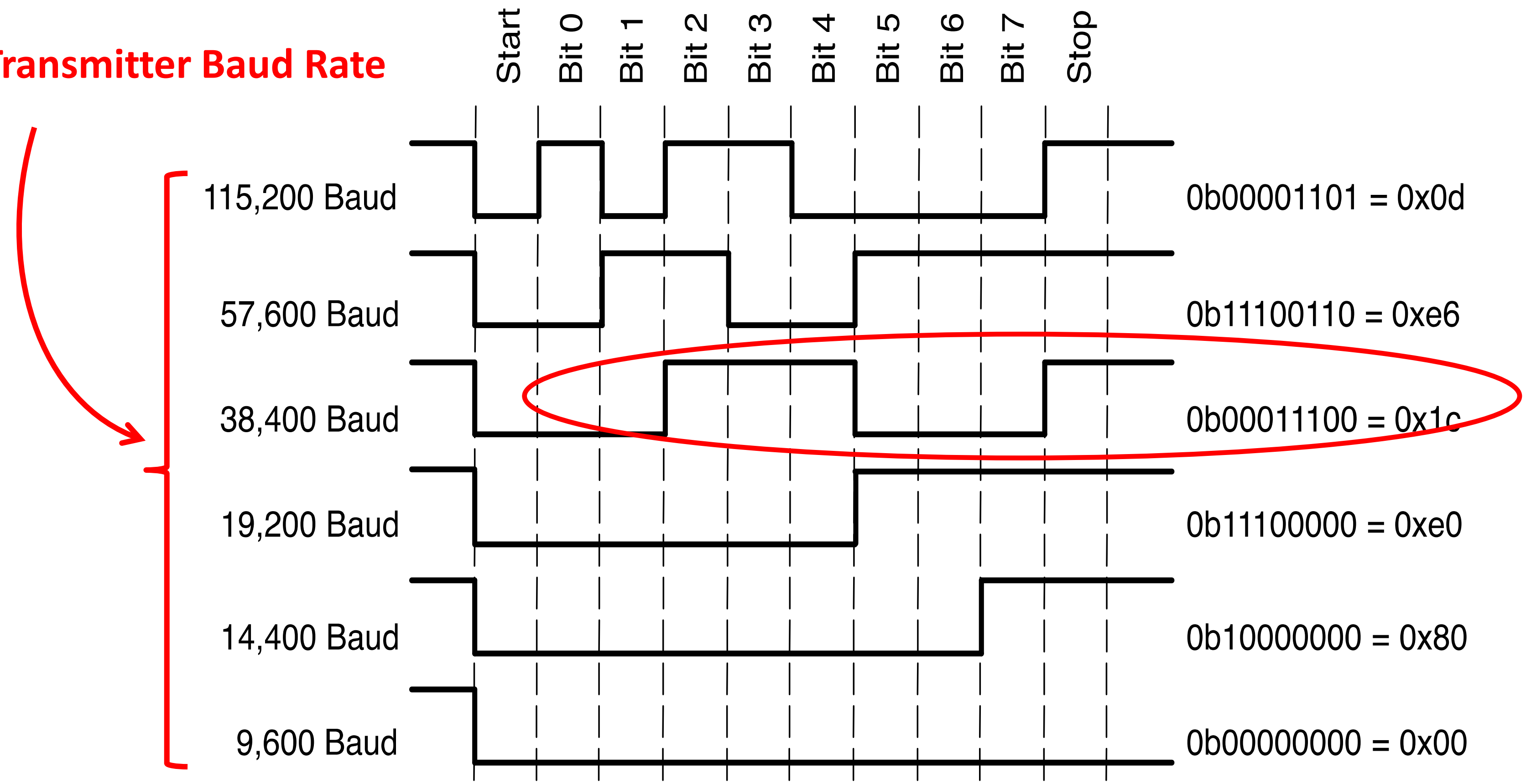
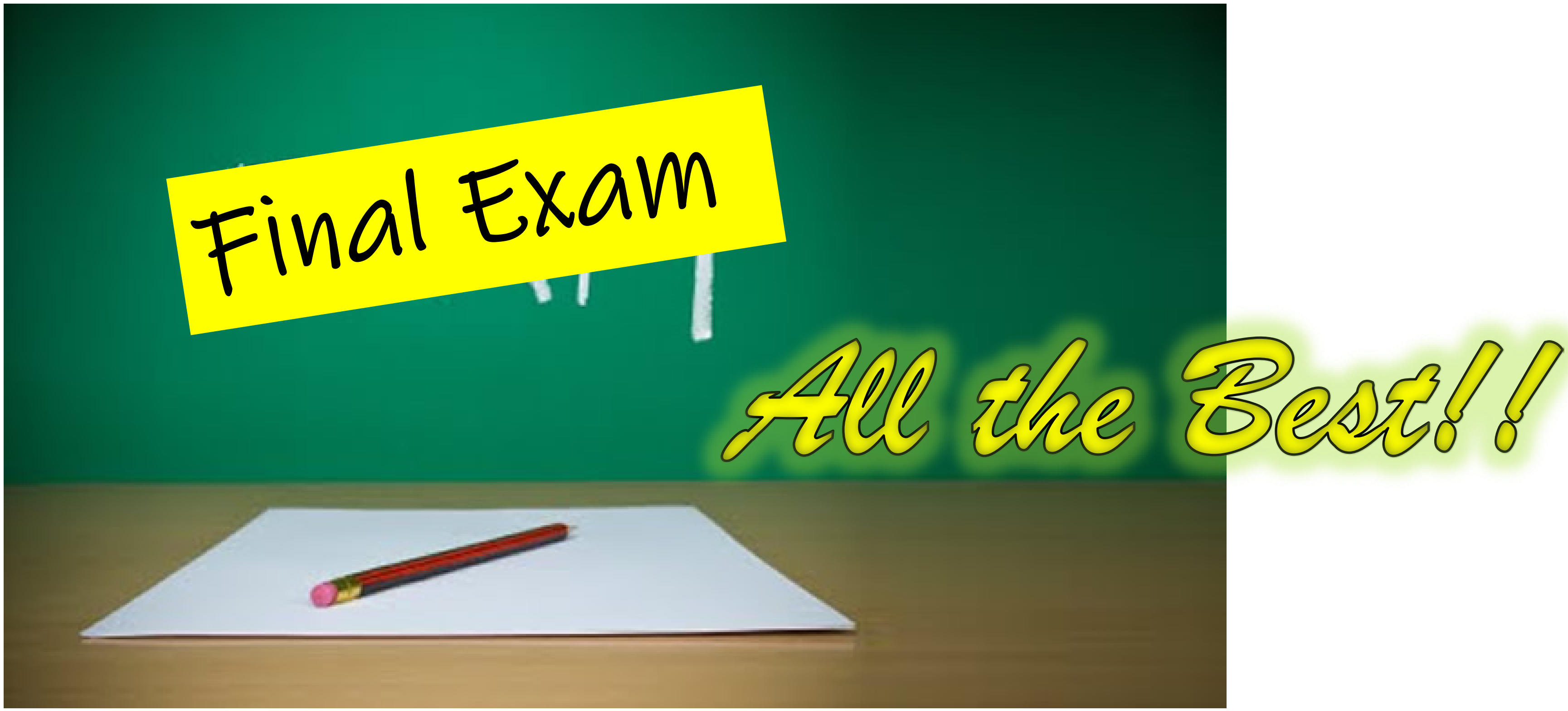


Figure 3. Bit Patterns for Baud Rates Between 115K and 9,600

If the receiving device knows that the transmitting device sends a 0x0D, then it'll be able to derive the baud rate based on the value sampled.

7. Why does the Processor Cx1006-200M16 has two different types of non-volatile memory (Flash and EEPROM) on-chip?

- Flash has a larger page size (order of **Kbytes** usually) so it can only be erased at Kbytes level. This is not suitable for certain use case such as storing of user configuration parameters, which is typically done at order of **bytes** level. An EEPROM, which has a smaller page size (tens of bytes) is more suitable for such use case as it result in less page erases. Also, EEPROM has a larger erase cycle endurance i.e. it could **tolerate a larger number of erasures** before failing.



Help?

- Q/A session : On site/line (this Friday)
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ANY QUESTIONS?